

FULL MARKS: 20

2. Design a combinational circuit using comparator ICs and logic gates to determine the **largest among three input numbers**: A , B , and C . The circuit must satisfy the following specifications: 7 +
1 + 2
(CO5)
(PO3)

- **Inputs:** Three binary numbers A , B , and C (you are not required to consider the number of bits, assume they are compatible with the comparator IC).
- **Outputs:** Three output signals:
 - **A largest:** High (1) if $A > B$ and $A > C$
 - **B largest:** High (1) if $B > A$ and $B > C$
 - **C largest:** High (1) if $C > A$ and $C > B$
- It is guaranteed that no two inputs are equal. Hence, exactly one of the outputs will be high at a time.
- **Comparator IC:** You may only use 2-input comparator blocks. Each comparator has two n -bit number inputs (X and Y) and three outputs: $X > Y$, $X < Y$, and $X = Y$, where X and Y are two n -bit numbers.
You are not required to design the comparator IC, it may be represented as a block in your diagram.
- **Design Requirement:** Using these comparator IC blocks and any other component as you wish, draw the **block diagram** of a combinational circuit that produces the required outputs. You may assume ideal behavior and unlimited availability of logic gates and comparators.

Note: A block diagram means that you do not need to draw the internal structure of the comparator ICs. Instead, use labeled blocks showing inputs and outputs clearly.

Answer the following questions -

- a) Design the above-mentioned circuit.
- b) Explicitly mention the number of 2-number comparator ICs you have used.
- c) Briefly explain the working mechanism (avoid unnecessary details).
- d) **Bonus 5 Marks** for proposing the most efficient solution.

3. Given,

3.5 + 2

$$F_1(A, B, C) = \sum m(4, 6, 7)$$

(CO5)
(PO3)

$$F_2(A, B, C) = \sum m(1, 5, 6)$$

Complete the PLD circuit for F_1 and F_2 by making the necessary connections (using 'X' at junctions) using the following **PLA (Programmable Logic Array)** structure.

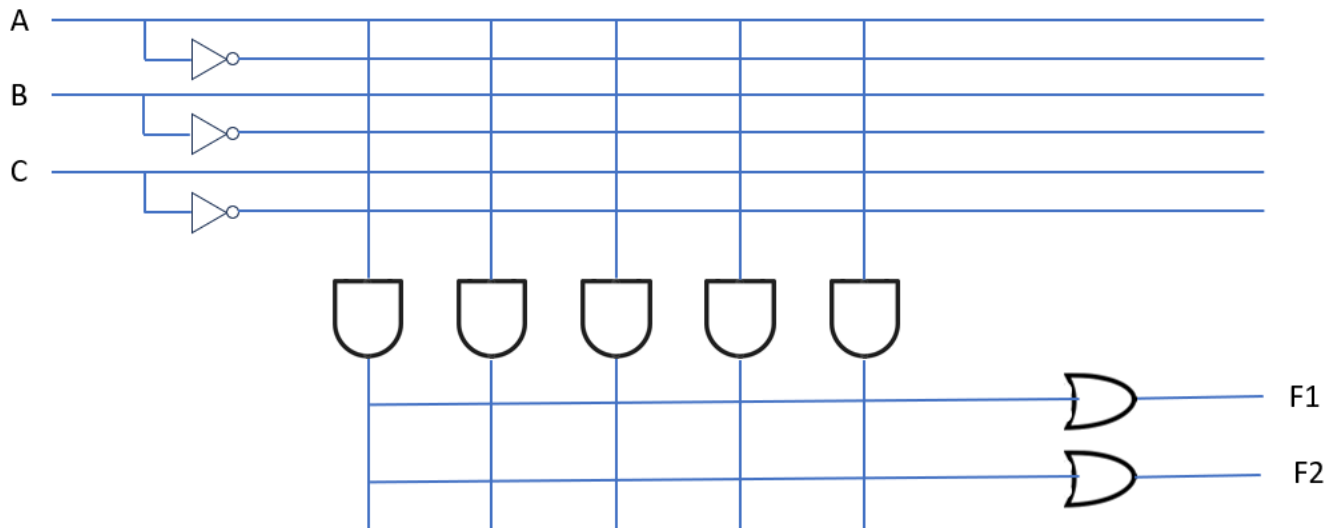


Figure 1: PLA Circuit

And answer the following questions-

a) A PLA circuit structure is defined by $n \times p \times m$. Write down what each symbol represents and provide their values according to the given PLA circuit diagram that you have constructed.

- n : _____, Value: _____
- p : _____, Value: _____
- m : _____, Value: _____

b) Put a tick next to the correct option.

For a **PAL (Programmable Array Logic)**:

- AND Array: ☐ Fixed ☐ Programmable
- OR Array: ☐ Fixed ☐ Programmable